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Title: Artificial Intelligence Auditability and Auditor Readiness for Auditing Artificial Intelligence Systems

IEEE Citation

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Summary:

This paper addresses the growing need for auditing artificial intelligence (AI) systems amid their rapid adoption in business, highlighting challenges like biases, lack of transparency, privacy erosion, and ethical risks that could lead to discrimination or human rights issues. The authors argue that while financial and information system audits are well-established, AI audits remain underdeveloped, leaving potential harms unaddressed. Drawing from semi-structured interviews with 23 AI professionals experienced in audits, the study proposes a comprehensive framework for AI auditability, mapping key dimensions such as fairness, explainability, robustness, and compliance to AI system components like data, models, and processes. It emphasizes that AI audits require multidisciplinary teams, including AI engineers, data scientists, IT auditors, domain experts, and ethicists, to ensure systems are lawful, ethical, and reliable.

The methodology focuses on qualitative insights from practitioners, verifying literature-based auditability measures and identifying auditor competencies, such as technical skills in machine learning, ethical knowledge, and governance expertise. Key findings reveal gaps in current practices, including insufficient AI governance standards, explainability issues in "black box" models, and poor quality controls for training data. The framework outlines auditability measures for each AI lifecycle stage design, development, deployment, and monitoring to mitigate risks like algorithmic bias or privacy violations. It also extends benefits to financial audits involving AI, helping prevent ethical pitfalls in accounting systems.

In conclusion, the authors position this as foundational research in a nascent field, urging the audit profession to build competencies through training and collaboration. By implementing the proposed framework, AI developers can enhance system trustworthiness, while auditors gain tools to conduct high-quality reviews. The paper calls for further studies to refine AI audit standards, contributing to safer AI integration across industries and informing policy on responsible AI use.